

Utilizando Comparable

```
import java.util.*;

class Student implements Comparable<Student> {
    String name;
    int grade;

    Student(String name, int grade) {
        this.name = name;
        this.grade = grade;
    }

    // comparacion dentro de la clase
    @Override
    public int compareTo(Student other) {
        return this.grade - other.grade; //compara notas
    }

    @Override
    public String toString() {
        return name + " - " + grade;
    }
}

public class EjemploComparable {
    public static void main(String[] args) {

        ArrayList<Student> list = new ArrayList<>();

        list.add(new Student("Ana", 90));
        list.add(new Student("Luis", 75));
        list.add(new Student("Marcos", 68));
        list.add(new Student("Maria", 87));

        Collections.sort(list);

        System.out.println("Sorted by grade:");

        for (Student s : list) {
            System.out.println(s);
        }
    }
}
```

Utilizando Comparator

```
import java.util.*;

class Student {
    String name;
    int grade;

    Student(String name, int grade) {
        this.name = name;
        this.grade = grade;
    }

    @Override
    public String toString() {
        return name + " - " + grade;
    }
}

// Clase comparator por separado
class NameComparator implements Comparator<Student> {

    @Override
    public int compare(Student s1, Student s2) {
        return s1.name.compareTo(s2.name); //compara nombres
    }
}

public class ComparatorExample {
    public static void main(String[] args) {

        ArrayList<Student> list = new ArrayList<>();

        list.add(new Student("Ana", 90));
        list.add(new Student("Luis", 75));
        list.add(new Student("Marcos", 68));
        list.add(new Student("Maria", 87));

        Collections.sort(list, new NameComparator());

        System.out.println("Ordenado por nombre:");

        for (Student s : list) {
            System.out.println(s);
        }
    }
}
```